

What the 010 Trigenic Model Is Measured Against

Additive nulls, what each model actually sees, and whether the comparison is fair

Michael Volk

September 9, 2026

Abstract

A model that claims to learn genetic interaction has to beat a null that cannot represent interaction. Fitting that null on experiment 010 puts a per-gene additive ridge at 0.400 test Pearson against the cell graph transformer’s 0.438 to 0.455, and puts a model that sees only which recurring gene pair a triple contains, ignoring the third gene entirely, at 0.390. Reading the model code establishes what the comparison is between: the transformer’s encoder never sees the strain, the nine interaction graphs enter training through a penalty on layer-1 attention rather than through the function that maps a checkpoint to a prediction, and the strain-dependent computation is a set function over the three perturbed gene identities, which is the same input and the same hypothesis-class shape the nulls have. That penalty was not incidental: it carried 99.99 percent of the training loss, the regularized heads recover their graphs at 0.73 to 1.00 recall, and a matched pair of runs with the penalty switched off fails to train at all. Rebuilding the forward pass reproduces the recorded metrics to the fourth decimal and yields per-record predictions. Those show the transformer is not a relabeled additive model, correlating with it at 0.73 to 0.76 rather than the 0.999 reported for MULTI-evolve, and that its gain over the null is +0.04 to +0.055 Pearson with bootstrap intervals excluding zero. Residualizing each prediction on the additive model isolates the non-additive content, which is 42 to 47 percent of prediction variance, agrees between two training runs at 0.53 to 0.58, and tracks the part of the label the additive model misses at 0.23 to 0.25.

The result that changes what the metric means is the split. Every record in this build carries one of 420 Kuzmin query doubles, and the published split is random over records, so the same query double appears in train and test. Refitting on a query-pair-disjoint split collapses the additive null from 0.400 to 0.127 ± 0.033 across five folds and inverts the ladder, with the nonlinear baseline falling below the additive one. Separately, the predictions over the unmeasured design space that produced the panel selection are invalid: the genome gene set shrank from 6,607 to 6,579 between inference runs, and the 28 missing mitochondrial open reading frames sort first, so every gene index was shifted by 28 and each triple was scored as a different triple.

Section status:  ready  author-reviewed, keep stable  not done, agents may edit freely

Provenance:  not in the library mirror; verify at the linked source  from a review’s summary table, not the primary paper

Contents

1	The question 	3
2	What is being predicted 	3
2.1	The label 	3
2.2	The build 	3
2.3	The screen design, which is the confounder 	4
3	What each model actually sees 	4
3.1	The transformer’s per-sample input 	4
3.2	The encoder does not depend on the strain 	4
3.3	Where the nine graphs act, and where they do not 	5
3.4	The transformer is a set function on three genes 	5
4	The models 	6
4.1	Where each baseline comes from 	6
4.2	Notation and the interaction operator 	7
4.3	B0, the train mean 	7
4.4	B1, the additive per-gene ridge 	7
4.5	B2, additive plus recurring gene pairs 	7
4.6	B3, the empirical screen mean 	8
4.7	B4, the screen indicator 	8
4.8	B5, nonlinearity on the same features 	8
4.9	Summary of capacity 	9
5	Is the comparison fair 	9
5.1	Which split the reported numbers come from 	9

5.2	What the transformer metrics are	10
5.3	What each model sees	10
5.4	Where the comparison is not clean	10
5.5	Model size	11
6	Held-out results	11
6.1	Reading the ladder	11
6.2	Verifying the architecture reading	12
6.3	What the graphs did during training	13
6.4	Record-by-record agreement	14
6.5	The query-pair-disjoint split	15
6.6	The same ladder on the replacement build	16
7	The design space, and a defect found there	18
7.1	The comparison as run	18
7.2	Why, and what it leaves standing	18
7.3	Seed instability in the selection tail, separately	18
8	Modernizing the 010 build	19
8.1	What actually broke	19
8.2	The migration, and why it copies rather than rebuilds	19
8.3	Which experiment ran on which data	19
9	What this establishes, and what it does not	20
9.1	Established	20
9.2	Not established	20
9.3	Next	21

1 The question ✗

Tran et al. introduced MULTI-evolve in *Science* in 2026 (10.1126/science.aea1820), a protein engineering framework whose fully connected networks were said to “learn the epistatic landscape” from single and double mutants and to identify synergistic combinations at higher order. Visani, Verma and DeWitt replied in a preprint, *Additive baselines furnish no evidence for epistasis learning by MULTI-evolve* (bioRxiv 2026.04.23.719915), which does one thing: it fits the null that the original paper never fit.

An **additive model** over mutations predicts a phenotype as a sum of per-mutation terms. By construction it assigns exactly zero to every interaction. It is therefore the natural null for any claim that a model has learned interaction, in the same way that an intercept-only model is the null for a claim that a model has learned anything.

Their result is that the network’s predictions across the combinatorial variant space correlate with a ridge-regularized additive model’s at $r > 0.999$ for all three engineering campaigns, that the variants selected for construction are ranked nearly identically by both, and that on held-out test data the network does not significantly outperform either the additive model or a zero-hidden-layer version of its own architecture. The engineering result is real; the attribution to epistasis learning is not supported.

The analogous question for experiment 010 is whether the equivariant cell graph transformer’s held-out performance on the trigenic interaction score survives the same test. Traditional machine-learning baselines exist in this repository for the mixed digenic and trigenic set under `experiments/002-dmi-tmi` and for fitness under `experiments/smf-dmf-tmf-001`. None had been fit on the 010 build, which is triples only.

One difference in the setup, and it cuts against the null. In the protein setting the quantity predicted is the multimutant phenotype itself, which per-mutation effects explain directly. In 010 the quantity predicted is already a residual. The trigenic interaction score subtracts the expectation built from single and double mutant fitness before the model ever sees it, so the additive part of fitness has been removed by construction. An additive-in-single-gene-effects model is therefore a harder null here than it is there, and its performance is more surprising, not less.

That argument has one loophole, and quantifying it turns out to matter more than anything else in this document. It is taken up in Section 6 as baseline B4.

2 What is being predicted ✗

2.1 The label ✗

Write f_i for the fitness of the single mutant Δi , f_{ij} for the double, f_{ijk} for the triple, each relative to wild type. The **digenic interaction score** is the departure of the double from the multiplicative expectation,

$$\varepsilon_{ij} = f_{ij} - f_i f_j, \quad (1)$$

and the **trigenic interaction score** is the departure of the triple from the expectation built out of the singles *and* the doubles,

$$\tau_{ijk} = f_{ijk} - f_i f_j f_k - f_i \varepsilon_{jk} - f_j \varepsilon_{ik} - f_k \varepsilon_{ij}. \quad (2)$$

▲ [external: Kuzmin et al. 2018, 10.1126/science.aao1729; the published definition of the released adjusted score column, not re-derived here]

Two properties of Eq. 2 matter for everything that follows. It is a *residual*: the additive-on-the-log-scale, multiplicative-on-the-linear-scale part of fitness is subtracted before the number is reported. And it is a *third-order* residual: the pairwise terms are subtracted too, so τ is by construction the part of the triple mutant phenotype that neither single nor double mutant measurements predict.

The 010 loader consumes the column **Adjusted genetic interaction score (epsilon or tau)** from the Kuzmin releases, which for a trigenic record is τ , and stores it under the label key `gene_interaction`.

2.2 The build ✗

The build is triples only, so there are no singles or doubles in it. The MULTI-evolve experiment of training on low order and extrapolating to high order is therefore not available here, and the question is the narrower one: given a training set of triples, how much of the held-out τ is recoverable without representing interaction.

Records	376,732
Perturbation count	3 for every record
Distinct perturbed genes	4,352
Sources	TmiKuzmin2018Dataset, TmiKuzmin2020Dataset
Label	gene_interaction (τ)
Split, seed 42	train 301,386 / val 37,673 / test 37,673
Train label SD	0.063186
Val label SD	0.062914
Test label SD	0.064187

Table 1: The 010 build. Every baseline and the transformer use these records, this split and this label. Read from `001-small-build/processed/label_df.parquet`, `is_any_perturbed_gene_index.json` and `data_module_cache/index_seed_42.json`.

2.3 The screen design, which is the confounder $\times$

Kuzmin’s trigenic screen crosses a **query** double mutant against an array of single mutants, so a triple is a query pair plus one array gene, and the same query pair recurs across hundreds or thousands of triples. Counting unordered gene pairs in the training split finds 593,978 distinct pairs, of which only 420 occur five or more times, the most frequent occurring 3,308 times. That concentration is the query-pair structure.

The 010 split is random over records, so a query pair that appears in training also appears in validation and test. Any model that can represent a per-pair offset can absorb whatever is common to a query batch, and none of that is trigenic biology. Baseline B4 in Section 4 is built to measure exactly this quantity.

3 What each model actually sees $\times$

The comparison is only meaningful if the models are given the same information. Reading the 010 code settles that, and the answer is stronger than expected: every model in this document, the transformer included, receives exactly one thing per record, the set of three perturbed genes.

3.1 The transformer’s per-sample input $\times$

The 010 dataset is built with the `Perturbation` graph processor. That processor writes the perturbed gene names, their indices, a boolean mask and the label into each sample, and nothing else (`graph_processor.py`, lines 1984 to 2058). There is no per-gene feature matrix: with an empty `node_embeddings` list in the config, the gene feature tensor of the cell graph has shape 6607×0 (`cell_data.py`, line 53). Gene identity is carried by a free lookup table of 6,607 rows and 180 columns (`equivariant_cell_graph_transformer.py`, line 1989), initialized randomly and fit by gradient descent.

No sequence, expression, ontology or other precomputed representation enters. The model is given gene identities, the same as a one-hot design matrix, and learns its own 180-dimensional vector for each.

3.2 The encoder does not depend on the strain $\times$

The forward pass builds its input as `gene_embedding(torch.arange(N))`, prepends a CLS token, and unsqueezes to shape $[1, N+1, 180]$. That leading dimension is 1, not the batch size. The eight transformer layers then run on this single tensor, which is assembled entirely from parameters and contains nothing from `batch`. Writing E for the embedding table and T for the eight-layer encoder,

$$Z = T(E) \in \mathbb{R}^{N \times 180}, \quad h_{\text{CLS}} \in \mathbb{R}^{180}, \quad (3)$$

and Z and h_{CLS} are the same for every record in the dataset. The repository states this itself, in the docstring of `PerturbationGraphPropagation`: “The encoder runs at batch 1 on the WILDTYPE graph, so `H_genes` and `h_CLS` are identical for every strain; the only strain-dependent step is `EquivariantPerturbationTransform`.”

So the eight encoder layers, which hold 3,129,120 of the model’s 4,774,861 parameters, perform no per-sample computation. They map one free parameter block to another. Whatever role they play, it is not to process the strain.

3.3 Where the nine graphs act, and where they do not ✗

The nine adjacency matrices enter the model in exactly one place. During training, a Kullback-Leibler penalty pulls one attention head of layer 1 toward each row-normalized adjacency $\tilde{A}^{(k)}$,

$$\mathcal{L}_{\text{graph}} = \sum_{k=1}^9 c_k \sum_{i=1}^N \sum_{j=1}^N \tilde{A}_{ij}^{(k)} \log \frac{\tilde{A}_{ij}^{(k)}}{\alpha_{ij}^{(k)}}, \quad \alpha^{(k)} = \text{softmax} \left(\frac{QK^\top}{\sqrt{d}} \right)_{h_k}, \quad (4)$$

and that is the only operation that reads them.

Not in the prediction. Three mechanisms could put adjacency into the computation that yields $\hat{y}$: a head mask on attention, the perturbation-propagation block, and an adjacency bias inside the attention logits. All three are off in the 010 configuration. Given a trained checkpoint, no adjacency tensor is an input to any operation between the gene indices and the prediction, and Section 6.2 checks this directly: setting the penalty weight to zero on a trained checkpoint moves the predictions by at most 1.3×10^{-6} .

In the weights. The penalty is computed inside the same forward call as the prediction, from the same pre-dropout attention tensor, and nothing detaches it. Its gradient reaches the layer-1 query and key projections, through them every layer-0 parameter, and through those the gene embedding table. So $Z = T(E)$ of Eq. 3, the per-gene vector the prediction is read from, is shaped by the nine graphs during training even though its value at inference comes off fixed weights. The graphs act on the parameters, not on the function from checkpoint to prediction. An ablation that zeroes the coefficient on a trained checkpoint tests only the second.

How large the penalty was. The per-graph weight c_k in Eq. 4 is a product of three factors: the per-head $\lambda_k = 10^{-3}$, a global scale of 1.0 read from `regression_task.loss.graph_regularization.lambda`, and an edge normalization meant to divide by the mean degree. That last factor is computed as `len(A.nonzero()[0])` summed over the nine matrices. On a dense torch tensor `A.nonzero()` has shape `[nnz, 2]`, so `[0]` is one index pair of length two, and the sum over nine graphs is 18 rather than the 2.5 million edges intended. Dividing by 18/6607 multiplies by 367.06,

$$c_k = 10^{-3} \times 1.0 \times \frac{6607}{18} = 0.367, \quad (5)$$

applied to a divergence summed over all 6,607 rows. The runs' logs record the consequence: against a point loss of 0.81 the graph term is 5,740, and the trainer's diagnostic `norm_weighted_graph_reg`, the graph term divided by the total loss, reads 0.99986 for all three checkpoints. The optimizer spent 99.99 percent of its objective on the graph penalty. Figure 1 shows the three loss terms and that share over the whole of training. Section 6.3 reports what the penalty bought: the regularized heads recover their graphs, and the matched runs without the penalty do not train.

The dominance was not what produced the accuracy. The 025 replication of this configuration (job 1598, Section 6.6) ran after the edge normalization had been corrected, so its graph term is 0.34 against a point loss of 0.80, a share of 0.29 to 0.33 rather than 0.9999, and it reached validation Pearson 0.4463 at epoch 14 against 0.4472 to 0.4619 for the three 010 checkpoints.

All nine heads carried the penalty. The configuration names the first two graphs `physical` and `regulatory`, which is what the cell graph called them at the December 2025 commit these runs used. The suffixed names, `physical_interaction` and `regulatory_interaction`, date from 2026-06-27. The runs log the edge-recovery keys `physical_L1_H0` and `regulatory_L1_H1`, which the trainer emits only for a graph it matched. An earlier version of this document claimed those two heads were unregularized and that the weight was squared to 10^{-6} ; both claims were checked against the run logs and are false for these runs.

3.4 The transformer is a set function on three genes ✗

What remains after Eq. 3 is the strain-dependent part. For a record with perturbed set S , $|S| = 3$, the perturbation transform performs cross-attention in which all N genes query the $|S|$ rows $\{Z_p : p \in S\}$, followed by a residual, a layer norm and a feedforward block, giving $H_i^{\text{pert}} = g(Z_i + c_S)$ where c_S is the attended context. The head then pools over the perturbed genes and reads out,

$$z_S = \frac{1}{|S|} \sum_{i \in S} H_i^{\text{pert}}, \quad \hat{y} = \text{MLP}([h_{\text{CLS}} \parallel z_S]), \quad (6)$$

with the MLP being `Linear(360, 180) → ReLU → Linear(180, 1)`.

Since h_{CLS} is constant, Eq. 6 is a function of z_S alone, and z_S is a permutation-invariant pooling of quantities derived from $\{Z_p : p \in S\}$. The 010 transformer is therefore a set function on the three perturbed gene identities, built on a

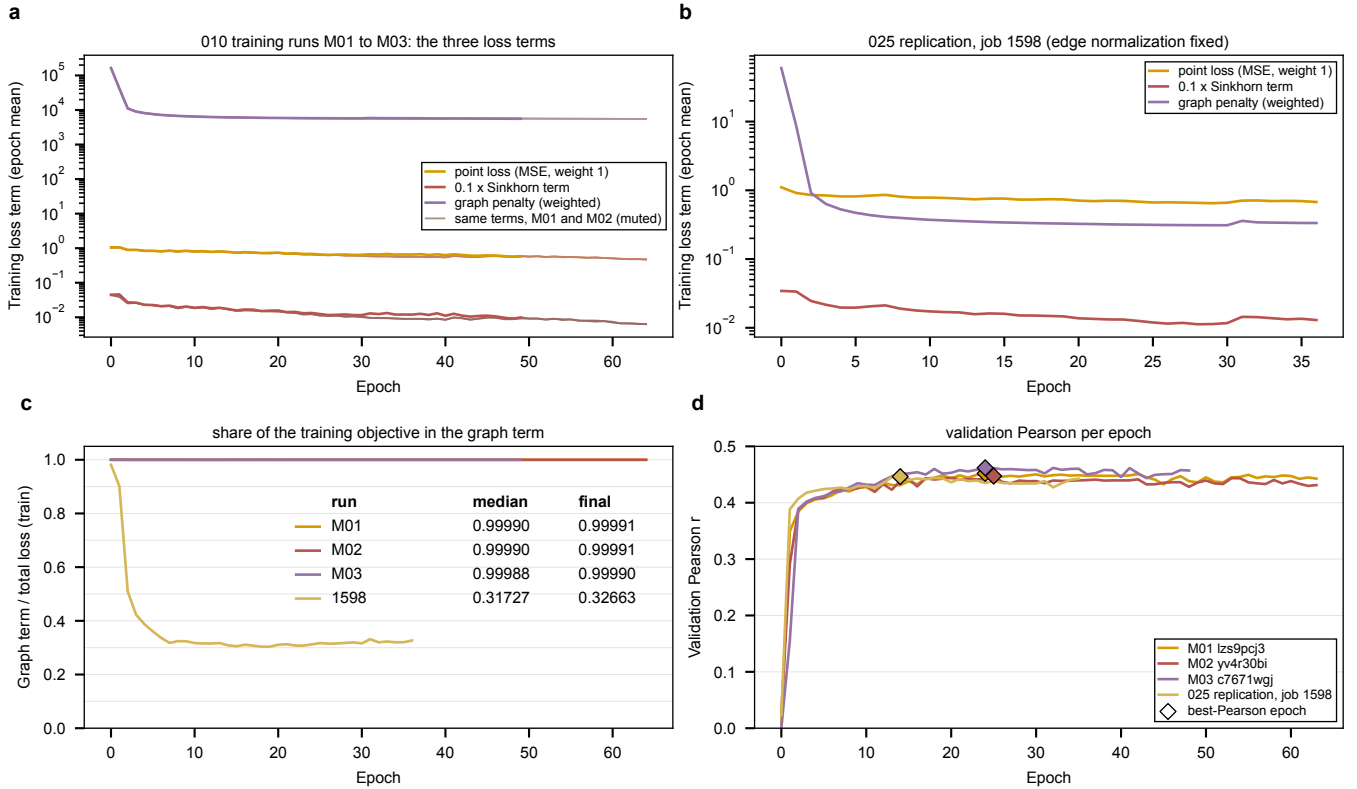


Figure 1: The training objective of the three 010 runs and of the 025 replication, per epoch. **a**, the point term, the Sinkhorn term at its weight of 0.1, and the weighted graph penalty for the 010 runs, on a log axis: the penalty sits four orders of magnitude above the point loss from epoch 5 on. **b**, the same three terms for job 1598, where the corrected edge normalization puts the penalty below the point loss. **c**, the graph term’s share of the total loss, recomputed from the logged terms: 0.9999 throughout for the 010 runs, 0.32 for the replication. **d**, validation Pearson per epoch with the best-Pearson epoch marked; the replication reaches the 010 band with the penalty at a third of the objective.

free per-gene vector. That is the same functional shape as the additive nulls, which are set functions on the same three identities built on a free per-gene scalar. The models differ in how expressive the set function is, not in what they are allowed to see.

This is what makes the comparison in Section 6 fair. It does not settle where the transformer’s advantage over the nonlinear baseline comes from. An earlier version of the working note assigned that gap to the nine interaction graphs, and a later one assigned it to capacity on the grounds that the graphs are absent from the prediction. Both overreached. The graphs shaped the weights this set function is built from, and Section 6.3 measures how heavily.

4 The models $\times$

4.1 Where each baseline comes from $\times$

Two of the six baselines are taken from the Visani preprint. The rest were built here for a confounder that the protein setting does not have. Naming which is which matters, because a baseline borrowed from a published critique carries that critique’s authority and one invented here does not.

Model	Origin	Corresponds to
B0	standard	intercept-only null
B1	Visani et al. 2026	their ridge-regularized additive model
B2	this document	B1 extended to observed pairs
B3	this document	unregularized counterpart of B4
B4	this document	the Kuzmin screen-structure confounder
B5	Visani et al. 2026	their zero-hidden-layer control, run in reverse

Table 2: Provenance of the six baselines.

The two external rows come from Visani, Verma and DeWitt, *Additive baselines furnish no evidence for epistasis*

learning by *MULTI-evolve*, bioRxiv 2026.04.23.719915 (<https://doi.org/10.1101/2026.04.23.719915>), which is itself a reply to Tran et al., *Science* 2026, 10.1126/science.aea1820 (<https://doi.org/10.1126/science.aea1820>). B1 is their additive null transplanted onto gene deletions. B5 inverts their zero-hidden-layer control: where they removed nonlinearity from the published architecture, B5 adds the smallest amount of nonlinearity to the additive feature space, so the two bracket the same gap from opposite sides.

B3 and B4 have no counterpart in the preprint because the confounder they measure is specific to a synthetic genetic array. A protein engineering campaign has no analog of a query double mutant recurring across thousands of measured variants. B4 turns out to carry the result that matters most here, so its being ours rather than theirs is worth stating plainly.

4.2 Notation and the interaction operator ✗

Let G be the 4,352 genes appearing in the build and let record n have perturbed set $S_n \subset G$ with $|S_n| = 3$. Encode it as $x_n \in \{0, 1\}^{|G|}$ with $x_{n,g} = 1$ exactly when $g \in S_n$, so that $\|x_n\|_1 = 3$ for every record. Write y_n for the trigenic interaction score of Eq. 2.

A model is a set function $f : 2^G \rightarrow \mathbb{R}$. Its capacity to represent interaction is measured by finite differences. For distinct genes g, h, k and any S disjoint from them, define

$$\Delta_g f(S) = f(S \cup \{g\}) - f(S), \quad (7)$$

$$\Delta_{gh} f(S) = f(S \cup \{g, h\}) - f(S \cup \{g\}) - f(S \cup \{h\}) + f(S), \quad (8)$$

$$\Delta_{ghk} f(S) = \sum_{U \subseteq \{g, h, k\}} (-1)^{3-|U|} f(S \cup U). \quad (9)$$

$\Delta_{gh} f \equiv 0$ says the model cannot express any pairwise interaction; $\Delta_{ghk} f \equiv 0$ says it cannot express any three-way interaction. These are the properties that make a model a null, and they are structural, holding for every parameter setting rather than for a fitted one.

4.3 B0, the train mean ✗

$\hat{y} = \bar{y}_{\text{train}}$. Every finite difference vanishes. Its mean squared error is the label variance, and it fixes the zero point against which all reductions are read.

4.4 B1, the additive per-gene ridge ✗

$$\hat{y}(S) = \beta_0 + \sum_{g \in S} \beta_g, \quad \hat{\beta} = \arg \min_{\beta} \sum_{n \in \text{train}} (y_n - \beta_0 - x_n^\top \beta)^2 + \alpha \|\beta\|_2^2, \quad (10)$$

with β_0 unpenalized and set to $\bar{y}_{\text{train}}$. Writing X for the $n_{\text{train}} \times |G|$ matrix whose rows are the x_n and $\tilde{y} = y - \bar{y}_{\text{train}}$ for the centered labels, the minimizer is the ridge normal equation

$$\hat{\beta} = (X^\top X + \alpha I)^{-1} X^\top \tilde{y}, \quad (11)$$

which the script evaluates by LSQR on the augmented system $[X; \sqrt{\alpha} I] \beta = [\tilde{y}; 0]$ rather than by forming the inverse. Substituting Eq. 10 into Eq. 8 gives $\Delta_{gh} \hat{y} = 0$ and hence $\Delta_{ghk} \hat{y} = 0$ identically. B1 is the direct analog of the additive null in the Visani preprint.

The design is well conditioned despite having 4,352 coefficients: with three ones per row and 301,386 training rows there are 904,158 gene observations, a median of 9 per gene in the training split with a maximum of 3,430. The ridge penalty is chosen on validation over $\alpha \in \{10^{-2}, 10^{-1}, 1, 3, 10, 30, 100, 300, 1000\}$ and the selected value is then reported on test.

4.5 B2, additive plus recurring gene pairs ✗

Let P be the set of unordered gene pairs occurring at least five times in the training split, $|P| = 420$. Adding one coefficient per such pair,

$$\hat{y}(S) = \beta_0 + \sum_{g \in S} \beta_g + \sum_{\{g, h\} \subseteq S, \{g, h\} \in P} \gamma_{gh}, \quad (12)$$

gives $\Delta_{gh} \hat{y} = \gamma_{gh}$ for a pair in P and zero otherwise. B2 therefore has second-order capacity on observed pairs. It still has *no* third-order capacity: $\Delta_{ghk} \hat{y} = 0$ identically. Since the label is itself a third-order residual, B2 remains a null with respect to the quantity being predicted.

The fit is the same ridge as B1 on the wider design $X = [X_G \ X_P]$, where X_P is the $n_{\text{train}} \times |P|$ pair-incidence matrix, so $(\hat{\beta}, \hat{\gamma})$ is Eq. 11 with X replaced by $[X_G \ X_P]$ and one penalty α on both blocks. The pair block has a closed form of its own that says what B2 learns. Every record contains exactly one recurring pair (Section 6.5), so the stationarity condition of the ridge objective in γ_{gh} involves only that pair’s records, and

$$\hat{\gamma}_{gh} = \frac{\sum_{n \in \text{train}, \{g,h\} \subseteq S_n} (y_n - \beta_0 - x_n^\top \hat{\beta})}{n_{gh} + \alpha}, \quad (13)$$

with n_{gh} the number of training triples carrying the pair. Each pair coefficient is the mean residual of the additive model over that query double’s screen, shrunk toward zero by $\alpha/(n_{gh} + \alpha)$. With n_{gh} in the hundreds to thousands and the selected α in the units, the shrinkage is at most a few percent, so B2 is B1 plus one nearly unshrunk offset per screen.

4.6 B3, the empirical screen mean ✖

B3 has no fitted parameters. Its per-gene and per-pair quantities are means of the training label over the records that contain the gene or the pair,

$$\bar{y}_g = \frac{\sum_{n \in \text{train}} x_{n,g} y_n}{\sum_{n \in \text{train}} x_{n,g}}, \quad \bar{y}_{gh} = \frac{\sum_{n \in \text{train}} x_{n,g} x_{n,h} y_n}{\sum_{n \in \text{train}} x_{n,g} x_{n,h}}, \quad (14)$$

so $\bar{y}_g$ averages y over exactly the training triples in which gene g is one of the three deletions, and $\bar{y}_{gh}$ over those carrying both g and h . The prediction backs off through three levels,

$$\hat{y}(S) = \begin{cases} \text{mean}\{\bar{y}_{gh} : \{g,h\} \subseteq S, n_{gh} \geq 5\} & \text{if any pair in } S \text{ recurs,} \\ \text{mean}\{\bar{y}_g : g \in S, n_g \geq 5\} & \text{otherwise, if any gene in } S \text{ has 5 training records,} \\ \bar{y}_{\text{train}} & \text{otherwise,} \end{cases} \quad (15)$$

with n_g and n_{gh} the training counts. On the random split every test record carries a recurring pair, so the first case always fires and B3 is the per-screen mean of the label. On a query-pair-disjoint split no test record carries a recurring pair, the first case never fires, and B3 becomes the mean of its three gene means.

4.7 B4, the screen indicator ✖

B4 drops the per-gene terms from Eq. 12 and keeps only the pair terms,

$$\hat{y}(S) = \beta_0 + \sum_{\{g,h\} \subseteq S, \{g,h\} \in P} \gamma_{gh}, \quad \hat{\gamma}_{gh} = \frac{\sum_{n \in \text{train}, \{g,h\} \subseteq S_n} (y_n - \bar{y}_{\text{train}})}{n_{gh} + \alpha}, \quad (16)$$

where the second expression is Eq. 13 with no gene block. Because the recurring pairs are the Kuzmin query doubles, B4 is a per-screen offset: for a triple made of a query pair and an array gene, the array gene contributes nothing at all. Whatever B4 achieves is therefore an upper bound on how much of the metric is explained by knowing which screen a record came from.

Eq. 16 at $\alpha \rightarrow 0$ is $\bar{y}_{gh}$ of Eq. 14, so on the random split, where the first case of Eq. 15 always fires, B3 and B4 are the same predictor up to the shrinkage $\alpha/(n_{gh} + \alpha)$, which Pearson does not see. Table 6 reports them equal to three decimals for that reason. On a disjoint split they part company: B4 is structurally zero and B3 falls back to gene means.

4.8 B5, nonlinearity on the same features ✖

Give each gene a free vector $e_g \in \mathbb{R}^{128}$, sum over the perturbed set and read out with a two-hidden-layer network ϕ ,

$$\hat{y}(S) = \phi \left(\sum_{g \in S} e_g \right), \quad \phi : \mathbb{R}^{128} \rightarrow \mathbb{R}. \quad (17)$$

This is permutation invariant and uses precisely B1’s feature space, the identity of the three genes and nothing more. Because ϕ is nonlinear, $\Delta_{gh} \hat{y}$ and $\Delta_{ghk} \hat{y}$ are not identically zero, so B5 is not a null. It is the smallest departure from B1 that has interaction capacity, and it isolates what capacity alone buys.

Comparing Eq. 17 with Eq. 6 shows they are the same construction. Both give every gene a free vector, pool over the three perturbed genes in a permutation-invariant way, and pass the pooled vector through a small network. The transformer’s pooling is richer, since the context c_S is produced by attention over the perturbed set rather than by a plain sum, and its per-gene vectors are $Z = T(E)$ rather than a raw table. Trained with three seeds, B5 reports a spread rather than a single number.

4.9 Summary of capacity ✗

Model	Per-gene quantity	Set aggregation	Fit by	$\Delta_{gh} \neq 0$	$\Delta_{ghk} \neq 0$
B0	none	none	train mean	no	no
B1	free scalar β_g	sum	ridge, Eq. 11	no	no
B2	free scalar β_g	sum, plus γ_{gh} per recurring pair	ridge, Eq. 13	yes	no
B3, pair case	none	mean of $\bar{y}_{gh}$ over the recurring pairs in S	label means, Eq. 14	yes	no
B3, gene case	$\bar{y}_g$, the label mean over training triples containing g	mean over S	label means, Eq. 14	no	no
B4	none	γ_{gh} per recurring pair	ridge, Eq. 16	yes	no
B5	free 128-vector e_g	sum, then MLP	gradient descent	yes	yes
CGT	free 180-vector via $T(E)$	attention over S , mean, MLP	gradient descent	yes	yes

Table 3: What each model can represent. B3 is listed by the case of Eq. 15 that fires: the pair case on the random split, the gene case on a query-pair-disjoint split. The last two columns are structural properties of the hypothesis class, not fitted results. Only B5 and the cell graph transformer can express a three-way interaction, which is what the label is.

5 Is the comparison fair ✗

Six things have to match before a baseline number can sit in the same table as a transformer number. Table 4 names them and gives the verdict on each; the subsections that follow give the evidence.

#	Verdict	What has to match
1	matches	Records and split. All models use the same 376,732 records and the same <code>index_seed_42.json</code> partition.
2	matches	Label. All predict <code>gene_interaction</code> , the published adjusted τ of Eq. 2.
3	matches	Reported split. Test is reported for every row, and validation is what every row selected on.
4	matches	Metric definition and denominator. Pearson, Spearman and mean squared error over the full 37,673 test records in raw τ units for every row.
5	matches	Information available. The identity of the three perturbed genes, and nothing else, for every model (Section 3).
6	does not	Label normalization. The transformer trains on z-scored labels whose mean and standard deviation were computed once, before the split, from all 376,732 records; the baselines use the training mean only. Pearson is invariant to the two constants, so this cannot move the reported metric (Section 5.4).

Table 4: What has to match for the comparison to be fair. Row 6 is the one mismatch. It is a hygiene defect in the 010 runs rather than a source of advantage, and the 025 arms from the disjoint run onward fit on the training split.

The loss functions also differ between the baselines and the transformer, but that is a difference in what each model was trained to do rather than in how the comparison is scored, so it is not one of the six and is treated separately below.

5.1 Which split the reported numbers come from ✗

Both validation and test numbers are available for the transformer, and the table in Section 6 reports **test**. The distinction matters because the three transformer checkpoints are *best-Pearson* checkpoints, selected on validation

across roughly 25 to 64 validation epochs. A validation Pearson chosen that way is the maximum of many draws and is upward-biased as an estimate of capability; its test Pearson is not, because test was not consulted during selection.

The baselines follow the same discipline. B1, B2 and B4 select the ridge penalty on validation and report test. B5 early-stops on validation and reports test. So the test column compares like with like, and the validation column is biased in the same direction for every row.

5.2 What the transformer metrics are ✗

The metrics come from the three checkpoints' re-evaluation runs under `$DATA_ROOT/wandb-experiments`, read from `files/wandb-summary.json` rather than retyped. Two families of key exist and they are not interchangeable.

- `val/gene_interaction/{MSE,RMSE,Pearson}` and the `test/` equivalents are torchmetrics objects updated once per batch and computed at epoch end, so they cover the *full* 37,673 records of each split. They are computed after the inverse transform, in raw τ units.
- `val_sample/Spearman_target_0` and its test counterpart come from a separate path that collects samples for plotting and is capped by `plot_sample_ceiling`. In the evaluation configuration that ceiling is 10^6 , above the split size, so no truncation fires and the Spearman is also over the full split. In the *training* configuration the same ceiling is 10,000, so a Spearman quoted from a training run would be over a random 10,000 of 37,673. The numbers here are from the evaluation runs.
- `val/transformed/gene_interaction/*` are the same metrics in standardized units and are not used here.

The baselines compute Pearson, Spearman and mean squared error over the same full splits in raw τ units, so the units and the denominators agree.

5.3 What each model sees ✗

Established in Section 3: the perturbed gene set, and nothing else, for every model in the comparison. No baseline is handicapped by being denied a feature the transformer had, and the transformer is not credited with information it never received.

5.4 Where the comparison is not clean ✗

Label normalization: what it is, when it happens, and what it can move. The transformer is trained on the z-scored label $(y - \mu)/\sigma$, the standard strategy of `C00LabelNormalizationTransform`. The two constants are computed once, at start-up, from `dataset.label_df`, and the transform is built before the data module splits the records, so in 010 the population is every record in the build. The logged constants confirm it: $\mu = -0.008024324$ and $\sigma = 0.063263549$ are the all-record statistics to nine digits, where the training split alone gives -0.007688739 and 0.063186255 . The inverse transform is applied to every prediction before a metric is computed, so all reported numbers are in raw τ units.

What the choice of population can and cannot do. Pearson and Spearman are invariant to an affine map of the labels, so no choice of μ and σ changes the correlation a given model would reach; the two validation and test scalars that reach training this way carry no information a model could turn into held-out correlation. What the constants do set is the scale of the point loss relative to the two other terms of Eq. 18: a different σ multiplies the standardized mean squared error by $(\sigma_{\text{all}}/\sigma_{\text{train}})^2 = 1.0024$, a quarter of a percent, and a different μ is a constant offset the readout bias absorbs. Against a graph term that is 99.99 percent of the objective (Section 3.3) that is nothing. So row 6 of Table 4 is a discipline defect, held-out labels touching a training-time constant, and not a mechanism for the transformer's margin.

It is corrected from the disjoint run onward. The 025 trainer now names the fit population, `transforms.fit_on`, and the query-pair-disjoint arm fits on its 301,236 training records, where the constants are $\mu = -0.007843739$ and $\sigma = 0.063776418$; the 025 replication arm keeps `fit_on: subset` because reproducing 010 requires 010's constants. Both sets are written by `experiments/025-solid-growth/scripts/label_normalization_constants.py`.

The loss functions differ, deliberately. The baselines minimize squared error. The transformer minimizes

$$\mathcal{L} = \lambda_p \mathcal{L}_{\text{MSE}} + \lambda_d S_\epsilon \left(\frac{1}{B} \sum_b \delta_{\hat{y}_b}, \frac{1}{B} \sum_b \delta_{y_b} \right) + \lambda_g \mathcal{L}_{\text{graph}}, \quad (18)$$

with $\lambda_p = 1$, $\lambda_d = 0.1$, $\lambda_g = 1$. The middle term is a debiased Sinkhorn divergence with $p = 2$ and blur 0.05 between the empirical measure of the batch’s predictions and that of the batch’s targets, computed unpaired, which shapes the marginal distribution of the predictions rather than their per-record accuracy. The third term is the graph penalty of Section 3.3.

This is a difference in training objective, not in the evaluation, and it is part of what the transformer is. It is noted because a distributional term can move Pearson and mean squared error in opposite directions, so the two metrics should be read together rather than either alone.

Paired residuals had to be recomputed. The original evaluation runs wrote per-record prediction files to the cluster checkout, and those files are not on this machine. Recomputing them through the stock evaluation script fails on the frozen build, which no longer validates (Section 8). Section 6.2 gives the route around that, and Section 8 gives the migration that removes the obstacle for future work.

5.5 Model size ✗

Component	Parameters	Role
Gene embedding table	1,189,260	the only place gene identity lives
Encoder, 8 layers	3,129,120	strain-independent, see Sec. 3.2
Perturbation transform	391,140	the strain-dependent step
Readout head	65,161	pooling and MLP
CLS token	180	constant
Total	4,774,861	
B1 additive ridge	4,353	one scalar per gene, plus intercept
B5 embedding MLP	623,745	128 per gene, plus a two-layer network

Table 5: Parameter budget. Two thirds of the transformer sits in the encoder, which does no per-sample work, and a quarter in the embedding table. The strain-dependent computation is 456,301 parameters, under a tenth of the model.

6 Held-out results ✗

Model	Interaction capacity	Pearson	Spearman	MSE
B0 train mean	none	0.000	0.000	0.004124
B4 query pair only	pairwise, 420 pairs	0.390	0.362	0.003493
B3 hierarchical mean	pairwise, empirical	0.390	0.362	0.003493
B1 additive per-gene ridge	none	0.400	0.406	0.003460
B2 additive plus pair ridge	pairwise, 420 pairs	0.406	0.412	0.003442
B5 embedding MLP, seed 0	full	0.425	0.417	0.003382
B5 embedding MLP, seed 1	full	0.427	0.421	0.003395
B5 embedding MLP, seed 2	full	0.425	0.419	0.003381
CGT M01 lzs9pcj3	full	0.443	0.426	0.003400
CGT M02 yv4r30bi	full	0.438	0.424	0.003439
CGT M03 c7671wgj	full	0.455	0.426	0.003315

Table 6: Test split, 37,673 records, raw τ units. The interaction-capacity column repeats Table 3: only the last four rows can express a three-way interaction, which is what the label is. Ridge penalties are selected on validation; B5 early-stops on validation; the transformer checkpoints are selected on validation. Test was not consulted by any of them.

6.1 Reading the ladder ✗

The additive null is close, but the transformer does beat it. B1 reaches 0.400 against the transformer’s 0.438 to 0.455, so a model that cannot represent any interaction recovers 88 to 91 percent of the correlation. In explained variance the gap is $r^2 = 0.160$ against 0.207 for the best checkpoint, an increment of 0.047 of label variance. In error, the best checkpoint reduces mean squared error 19.6 percent below B0 and B1 already delivers 16.1 percent of that. This is not the DeWitt result, where the network collapsed onto the additive model at $r > 0.999$. The margin here is real. It is also small.

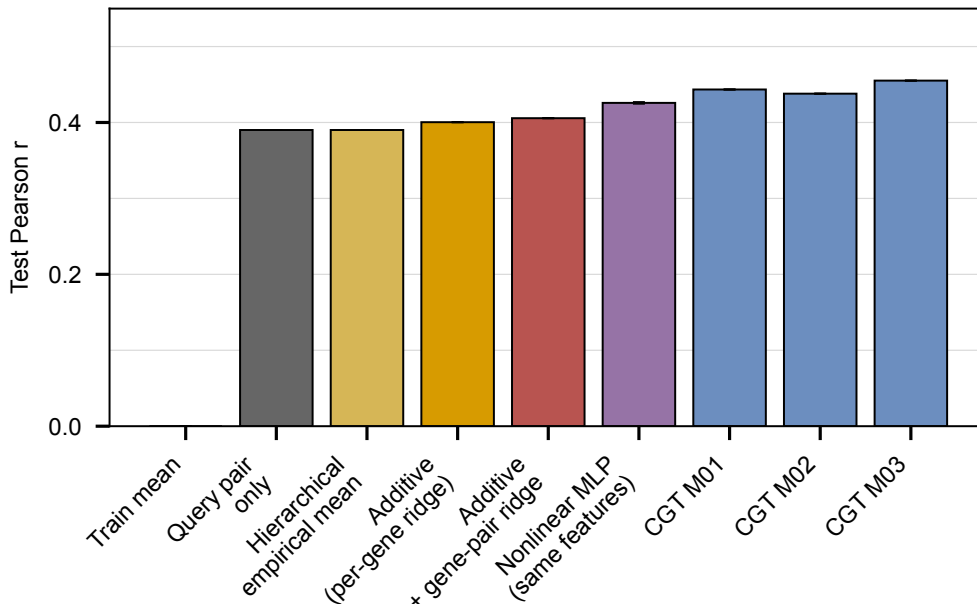


Figure 2: Test Pearson for every model. Bars carry an error bar only where the fit has a random component. B0 through B4 are closed-form solutions of a convex problem on a fixed split, so each is one deterministic number and has nothing to average over. The nonlinear baseline B5 is trained by stochastic gradient descent, and its bar is the mean of three seeds with the standard deviation across them. The three transformer bars are three separate training runs plotted individually rather than pooled, because they are not seed replicates: the script fixes the seed at 42, M01 and M02 differ only in run-to-run nondeterminism, and M03 also changes the scheduler’s first cycle length. Figure 5 is this ladder under the query-pair-disjoint split.

Seed spread is a large fraction of that margin. The three transformer runs span 0.438 to 0.455, a standard deviation of 0.0088. The three B5 seeds span 0.425 to 0.427, a standard deviation of 0.0011. So the transformer’s run-to-run spread is eight times the nonlinear baseline’s and is roughly a fifth of its own margin over B1. A single checkpoint’s number is not the capability.

Capacity closes about half the gap, and the graphs cannot be ruled out. B5 sees exactly B1’s features, has no graph and no attention, and reaches 0.426. That closes about half the distance from B1 to the transformer. On mean squared error B5’s 0.003382 is better than two of the three checkpoints.

An earlier version of this document attributed the whole remaining B5-to-transformer difference to capacity, on the grounds that the graphs are not in the forward pass. That inference does not hold, and Section 6.3 gives the measurements that break it. The graphs shape the trained weights even though they are absent from the prediction function, and the evidence that they shaped these weights heavily is direct. What can be said from the ladder alone is that richer set aggregation and eight times the parameters are worth at most 0.03 Pearson over B5, and that how much of that 0.03 is the graphs is not separable from these runs.

B4 is the result that changes what the metric means. B4 knows only which of 420 recurring gene pairs a triple contains. It never looks at the third gene. It reaches 0.390 test Pearson, 97 percent of B1’s and 86 percent of the best checkpoint’s. Those 420 pairs are the Kuzmin query doubles, the most frequent appearing 3,308 times, so B4 is close to a per-screen offset.

On a split that is random over records, then, most of what every model in Table 6 is scoring is the identity of the screen a triple came from. That is not trigenic biology and it would not survive a query-pair-disjoint split. It also explains why the additive null does so well on a label that was constructed to have the additive part removed: what B1’s per-gene coefficients absorb is largely query-pair structure reaching them through the genes that make up the query pairs.

Spearman compresses the ladder further. The rank correlations run 0.406 for B1, 0.412 for B2 and 0.417 to 0.421 for B5, against 0.424 to 0.426 for the transformer. The gap from the additive null to the transformer is 0.020 in Spearman against 0.045 in Pearson. Ranking is what a design-space selection consumes, so the smaller of the two is the more decision-relevant number.

6.2 Verifying the architecture reading x

Section 3 claims the encoder is strain-independent and that no adjacency tensor is an input to the prediction. Both are checkable rather than merely readable, and checking them also produces the per-record predictions that Section 9 lists

as missing.

The stock evaluation path cannot run against the frozen build, for the reasons Section 8 sets out. The route around it follows from the architecture. If the encoder really does not see the strain, then scoring a record needs only the perturbed gene indices, and the index space is the sorted base-graph node list, rebuildable from the genome without touching the LMDB. `score_010_checkpoints_directly.py` does that, loads each checkpoint into the model class, and recomputes the metrics.

The check is that the recomputed numbers must reproduce what the original evaluation runs recorded. A wrong gene ordering or a mis-loaded weight would give a plausible but wrong number, so agreement to several digits is the evidence that the reconstruction is faithful.

Checkpoint	Val Pearson		Val MSE	
	recomputed	recorded	recomputed	recorded
M01 lzs9pcj3	0.451970	0.451963	0.003244	0.003244
M02 yv4r30bi	0.447319	0.447155	0.003258	0.003259
M03 c7671wgj	0.461917	0.461881	0.003163	0.003163

Table 7: Scoring the checkpoints through a hand-built forward pass that computes the encoder once and feeds it only perturbed-gene indices reproduces the recorded metrics. The residual difference is at the fourth decimal and is consistent with the original runs using mixed bf16 precision.

Two details had to be right for this to work, and both are worth recording because getting either wrong produced a plausible but wrong answer. The index space is the sorted genome gene set, 6,607 genes, not the build’s own 6,579 gene set; using the latter gave a validation Pearson of -0.001 to 0.002 . And the readout pools by *mean*, whereas the current class defaults to *sum*; using the default gave 0.420 to 0.438, close enough to look credible and wrong.

That reproduction is the direct verification of Section 3.2. The encoder was evaluated exactly once, on a strain-independent input, and the recorded per-split metrics came back. If the encoder had needed the strain, this could not have happened.

The other claim needs a sharper statement than it was given. Setting the graph penalty weight from 1.0 to 0.0 on a trained checkpoint changes the maximum absolute test prediction by 9.0×10^{-7} , 1.2×10^{-6} and 1.3×10^{-6} , against a prediction spread of about 0.036. That establishes what it establishes: adjacency is not an input to the prediction function. It says nothing about whether the graphs shaped the weights being scored, and Section 6.3 shows they did.

6.3 What the graphs did during training ✗

Three independent measurements, none of which the inference-time ablation can see.

The penalty was almost the whole objective. The trainer logs the graph term, the point term and their normalized shares. Against a validation point loss of 0.81 the graph term is 5,740, and the share `norm_weighted_graph_reg` is 0.99986 for all three checkpoints. Eq. 5 accounts for the size: the intended division by mean degree is a multiplication by 367, and the resulting per-graph coefficient is 0.367 on a divergence summed over 6,607 rows.

The regularized heads recover their graphs. The trainer scores each regularized head by asking, for every gene i with degree d_i , what fraction of i ’s true neighbors fall in that head’s top d_i attention targets.

Graph	Head	M01	M02	M03
physical	0	0.940	0.940	0.938
regulatory	1	0.992	0.992	0.991
tflink	2	0.979	0.979	0.979
string neighborhood	3	0.952	0.951	0.950
string fusion	4	0.999	1.000	1.000
string cooccurrence	5	1.000	1.000	1.000
string coexpression	6	0.729	0.731	0.729
string experimental	7	0.774	0.772	0.773
string database	8	0.996	0.996	0.995

Table 8: Neighbor recall at each gene’s own degree, layer 1, one head per graph. Ordinary self-attention has no reason to place its top d_i mass on a gene’s true neighbors, so these values are attributable to the penalty of Eq. 4.

Removing the penalty stops the model training. A matched pair of 30-epoch configurations exists for exactly this test, identical but for the graph term’s weight. With the weight at 1.0, three replicates reach validation Pearson

0.464, 0.456 and 0.456. With it at 0.0, the three companion runs reach -0.007 , -0.031 and a diverged run that returned no number, with training Pearson around 0.002.

That is a strong result and it needs a careful reading. It establishes that the graph term is load-bearing for optimization in this configuration. It does not establish that the graphs supply biological information the model uses, because a term carrying 99.99 percent of the loss is also doing whatever conditioning, scaling and implicit regularization that share implies. Separating the two would need a control with a comparable penalty against permuted or degree-matched random graphs, which has not been run.

6.4 Record-by-record agreement ✗

With per-record predictions available, the comparison can be made the way the DeWitt preprint makes it, between the predictions themselves rather than between their summary statistics.

Model	Pearson with		
	CGT M01	CGT M02	CGT M03
B1 additive ridge	0.752	0.726	0.759
B2 additive plus pair	0.759	0.734	0.765
B4 query pair only	0.729	0.706	0.735
B5 embedding MLP	0.788	0.784	0.810
CGT M01		0.788	0.821
CGT M02	0.788		0.804

Table 9: Correlation between models’ test-set predictions, 37,673 records. The transformer does not reproduce the additive model, which is where 010 departs from the MULTI-evolve case. But two transformer training runs agree with each other at 0.788 to 0.821, barely more than a transformer agrees with the additive model at 0.726 to 0.759, and no more than it agrees with the nonlinear baseline at 0.784 to 0.810.

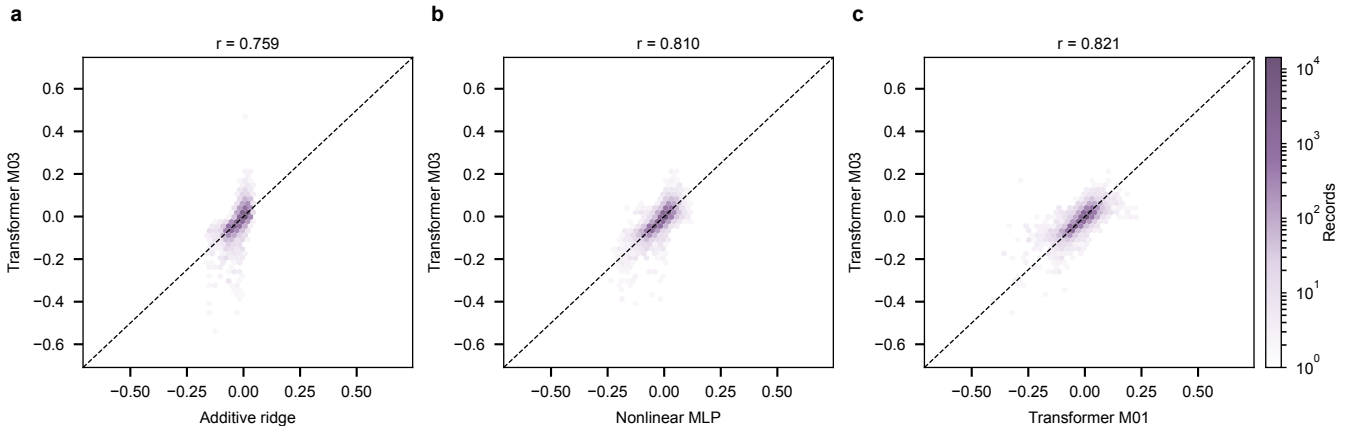


Figure 3: Test-set predictions of one model against another. The transformer is no closer to a second training run of itself than it is to a nonlinear baseline that has no graph and an eighth of the parameters.

The headline distinction from the preprint. DeWitt’s networks correlated with the additive model at $r > 0.999$, so the network *was* the additive model. Here the correlation is 0.73 to 0.76, so the 010 transformer is genuinely computing something the additive model does not. That conclusion should be stated plainly, because it is the opposite of the preprint’s finding and it is what the null was fit to test.

How reproducible the extra content is, measured directly. Run-to-run correlation between two transformer checkpoints is 0.79 to 0.82. An earlier version read that against the 0.73 to 0.76 agreement with the additive model and concluded the non-additive content was largely irreproducible. That comparison is not sound, because both correlations are dominated by the additive part that every model reproduces, so neither number is about the non-additive content at all.

The direct measurement regresses each model’s predictions on B1’s and correlates what is left. On that scale the non-additive component is 43.5, 47.3 and 42.4 percent of each checkpoint’s prediction variance, so it is a large share of what the model outputs. Between two checkpoints it correlates at 0.533, 0.584 and 0.564. Against the part of the label B1 also fails to explain it correlates at 0.236, 0.234 and 0.253, where B5’s non-additive component manages 0.165.

So the non-additive content is neither noise nor stable. About half of it is shared between two training runs, and the shared part carries real signal about the residual label. The three runs are also not seed replicates in the strict sense:

the training script fixes the seed at 42, M01 and M02 differ only in run-to-run nondeterminism, and M03 additionally changes the scheduler’s first cycle length. A designed seed sweep has not been run.

The models are wrong in the same places. Correlating the residuals $y - \hat{y}$ of B1 and each checkpoint gives 0.907, 0.897 and 0.918. Almost all of the error is shared, so the transformer is not fixing a class of records the additive model gets wrong; it is making a modest, broadly distributed improvement.

The advantage is real and bounded. A paired bootstrap over test records, 2,000 resamples, puts the Pearson gain over B1 at +0.043 with a 95 percent interval [+0.023, +0.063] for M01, +0.038 with [+0.019, +0.058] for M02, and +0.055 with [+0.034, +0.074] for M03. All three intervals exclude zero. B5 gains +0.025 with [+0.017, +0.033]. So the transformer beats a no-interaction null by a margin that is statistically established and is between two and six hundredths of a correlation.

Rankings diverge where selection happens. Among the 100 records each model calls most negative, B1 and M03 share 31, B5 and M03 share 45, and B1 and B5 share 46. At $K = 10,000$ all pairs exceed 0.91. The models agree about the bulk and disagree about the tail, which is the part a design-space selection reads.

Between two transformer checkpoints the overlap rises monotonically with K : 0.10 to 0.30 at $K = 10$, 0.39 to 0.47 at $K = 100$, 0.57 to 0.60 at $K = 1,000$, and 0.91 at $K = 10,000$. Reading the tail figure as a defect would be wrong. An extreme quantile is where the fitted surface is least constrained by data and where the label itself is noisiest, so two runs that agree about the bulk should be expected to disagree about the last hundred. The consequence for practice is unchanged, and it is a reason to ensemble rather than a reason to distrust the architecture.

Prediction spread. Test prediction standard deviations are 0.025 for B1, 0.029 for B5 and 0.036 to 0.038 for the checkpoints, against a test label standard deviation of 0.064. Every model is heavily shrunk toward the mean, as a squared-error fit on a noisy target should be.

6.5 The query-pair-disjoint split ✗

B4’s 0.390 says most of Table 6 could be screen structure. The measurement that settles it is a split in which no query double appears in more than one part, and it has now been run.

The screen structure is exact, not approximate. Counting unordered gene pairs over all 376,732 records finds 739,315 distinct pairs, of which 420 occur five or more times. Those 420 cover 376,733 pair-instances. That is one more than the record count, so every record contains exactly one of the 420 recurring pairs and a single record contains two. Every record’s most frequent pair occurs at least 200 times. The 420 pairs are the Kuzmin query doubles, and grouping records by them recovers the screen exactly rather than by a threshold that has to be defended.

One disjoint split, and then five. Assigning whole query pairs gives 285 train, 67 validation and 68 test pairs, or 80.03, 9.97 and 10.00 percent of records, with no query pair shared between any two parts. On that split the additive ridge reaches 0.174 test Pearson, against 0.135 on its own validation part. A 4-hundredth gap between two parts of one split is a warning that 68 query pairs is a small sample of screens, so the reported numbers below are five-fold cross-validation with the query pair as the grouping unit.

Model	Random split	Query-pair disjoint	
	test r	one split	5-fold mean $\pm$ sd
B0 train mean	0.000	0.000	0.000 $\pm$ 0.000
B4 query pair only	0.390	0.000	0.000 $\pm$ 0.000
B3 hierarchical mean	0.390	0.173	0.066 $\pm$ 0.039
B1 additive per-gene ridge	0.400	0.174	0.127 $\pm$ 0.033
B2 additive plus pair ridge	0.406	0.170	0.124 $\pm$ 0.034
B5 embedding MLP	0.426	0.150	0.058 $\pm$ 0.037
CGT	0.438 to 0.455	not measured, needs retraining	

Table 10: Held-out Pearson under the published random-over-records split and under a split that never shares a Kuzmin query double. The cross-validated column is the mean over five folds, each holding out a disjoint set of query pairs, with B5 additionally averaged over three seeds. B4 is structurally zero once query pairs are disjoint, because no test record carries a pair that has a coefficient.

Most of the additive null’s score was screen identity. B1 falls from 0.400 to 0.127 ± 0.033 . In explained variance that is r^2 from 0.160 to 0.016, so roughly nine tenths of what the additive null appeared to explain was carried by query doubles it shared with the training split. The fold-to-fold spread, 0.088 to 0.151, is wide enough that any single

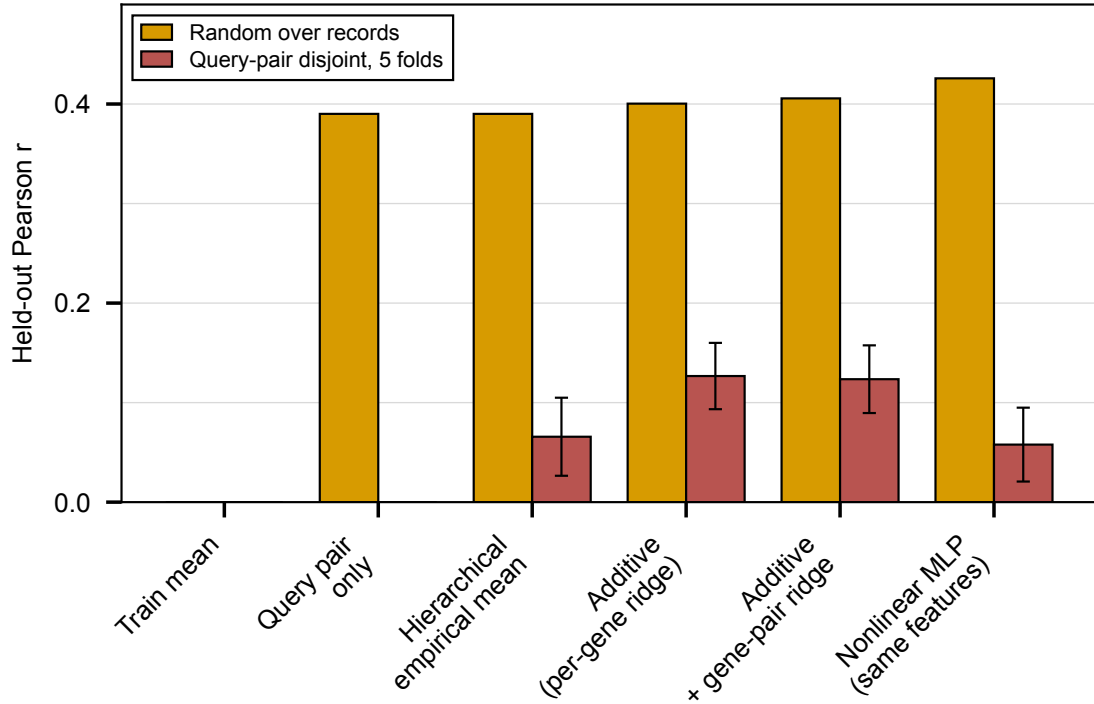


Figure 4: Each baseline under the published random-over-records split and under query-pair-disjoint folds. Error bars are the standard deviation across five folds. The query-pair-only model is structurally zero on the right, and the nonlinear model crosses below the additive one.

query-pair-disjoint split would misreport the number by several hundredths, which is why the single split in the middle column reads high.

The ladder inverts. Under the random split, interaction capacity bought a monotone climb from 0.390 to 0.455. Under the disjoint split the nonlinear baseline, which is the only baseline that can represent a three-way interaction, drops *below* the additive one, 0.058 ± 0.037 against 0.127 ± 0.033 . It also early-stops at epoch 3 to 7 rather than epoch 20 to 30, having reached training Pearson 0.43 while held-out performance was already falling. Generalizing to an unseen query double is a different problem from generalizing to an unseen record of a seen query double, and the extra capacity that helped on the second hurts on the first.

What is not measured. The transformer has no row in Table 10. Its checkpoints were trained on the random split, so every record of a disjoint test set was potentially in their training data, and scoring them there would measure memorization. A transformer number for this split requires retraining, which Section 8 makes possible and which is now queued on the replacement build (Section 6.6). Nothing here says the transformer’s margin over B1 does or does not survive; it says the quantity all the random-split rows were competing on was mostly screen identity.

6.6 The same ladder on the replacement build ✗

The 010 build is frozen and no longer loads (Section 8), so the transformer is retrained on experiment 025’s build, which holds the same 376,732 triples among 13.5 million records. Two arms train on those triples: arm R on 010’s own split, carried across by genotype identity, and arm Q on a query-pair-disjoint split of 331, 43 and 46 query pairs. Arm R has run (job 1598) and reproduces 010, validation Pearson 0.4463 at epoch 14 against 0.4472 to 0.4619 for the three 010 checkpoints. Arm Q is job 1609 and starts after this build. The baselines are refit on exactly the record pool and splits those two arms read, from the same committed index artifacts the trainer loads, so the transformer number from either arm has a null on the same data.

Arm R agrees because the data are the same, which the data show directly. Refitting is not how the two builds are checked against each other. Matching the 025 records to 010’s by genotype recovers all 376,732 of them, 376,526 labels are bit-identical, and over the remaining 206 the largest absolute difference is 5.6×10^{-17} at a Pearson of 1. The split is carried across by that same genotype identity. B0 through B4 are deterministic functions of the labels and the split, so their agreement to 3×10^{-13} in Table 11 follows from those two facts and measures floating point, not reproducibility. B5 is the one row where the builds can differ, and it differs by 0.006 against a three-seed standard deviation of 0.005 on the 025 fit. That run was on CPU where 010’s was on GPU, and the cause has not been traced.

What arm Q is asking. On the partition job 1609 will report on, the additive ridge reaches 0.185 and the nonlinear

Model	Arm R, random over records		Arm Q, query-pair disjoint
	010 build	025 build	025 build
B0 train mean	0.000	0.000	0.000
B1 additive per-gene ridge	0.400	0.400	0.185
B2 additive plus pair ridge	0.406	0.406	0.180
B3 hierarchical mean	0.390	0.390	0.142
B4 query pair only	0.390	0.390	0.000
B5 embedding MLP, 3 seeds	0.426 ± 0.001	0.432 ± 0.005	0.141 ± 0.007
CGT, three 010 checkpoints	0.438 to 0.455		
CGT, job 1598, val at best epoch		0.446	
CGT, job 1609			pending

Table 11: Test Pearson on the 025 build under the two splits the transformer arms use, beside the 010 numbers of Table 6. Arm R on the two builds agrees for every closed-form model to 3×10^{-13} , which follows from the record-level label and split identity rather than testing it; B5 is a stochastic fit and its run here was on CPU where 010’s was on GPU. Arm Q is one split, not the five folds of Table 10, and it is the exact partition job 1609 will report on. Job 1598’s number is a maximum over 36 validation epochs and has no test evaluation yet.

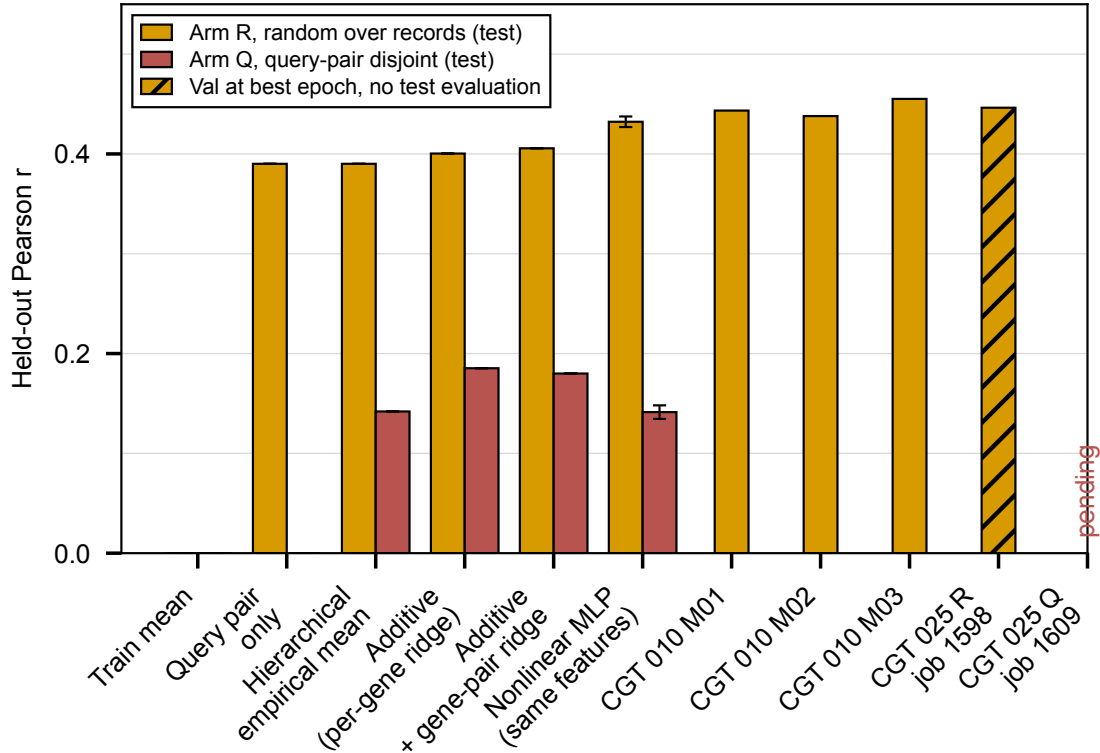


Figure 5: Held-out Pearson on the 025 build for each baseline under arm R and arm Q, with the transformer rows that exist: the three 010 checkpoints and the arm R replication. This is Figure 2 redrawn with the disjoint arm beside it. As there, only B5 carries an error bar, the standard deviation across its three seeds; B0 through B4 are deterministic fits on a fixed split. Each arm is one split and not the five folds of Figure 4, so the arm Q bars have no fold spread to show. The hatched bar is a validation maximum, not a test number, and the arm Q transformer slot waits on job 1609.

baseline 0.141, and B4 is structurally zero. A transformer number above 0.185 says the model carries something about held-out screens that per-gene additivity does not; one at or below it says the +0.04 margin of Table 6 was screen identity, like most of what the random split measured. The single-split reference here is higher than the five-fold mean of Table 10, 0.127 ± 0.033 , for the reason that table gives: one draw of 46 held-out screens is a small sample of screens, so the comparison to make is on this split, against these numbers, not against the fold mean.

7 The design space, and a defect found there ✗

Figure 1 of the Visani preprint is a scatter of network predictions against additive predictions across the whole combinatorial space, and its force comes from the near-identity of the two rankings: the variants selected for construction were recoverable without the network. The same comparison on 010 gave a result that turned out not to be about additivity at all.

7.1 The comparison as run ✗

Checkpoint `c7671wgj` scored three separate 010 inference spaces:

- `inference_1`, 4,370,595 triples over a 526-gene panel,
- `inference_2`, 479,195 triples,
- `inference_3`, 465,735,532 triples genome-wide, which is the space the panel-12 and panel-24 selections consumed.

Scoring the same triples with B1 and streaming the 4 GB prediction file gives, over the 325,631,845 triples of `inference_3` whose three genes all carry an additive coefficient, a Pearson correlation of 0.0018 and a top-100 overlap of zero. Stratifying by the training support of each triple’s least-observed gene does not account for it: even where all three genes have 800 or more training observations, $n = 867,669$, the correlation is 0.036.

On `inference_1` the same comparison behaves completely differently. B1 against the three checkpoints gives 0.548, 0.579 and 0.511.

7.2 Why, and what it leaves standing ✗

The model is a deterministic function of the perturbed gene set, so a triple scored twice by one checkpoint must receive the same number twice. Across these runs it did not, and the additive coefficient $\hat{\beta}$ of Eq. 10, fit on measured labels and keyed on systematic gene name, is what showed which run was wrong: a gene’s mean predicted effect tracks $\hat{\beta}_g$ at 0.58 on `inference_1` and at 0.05 and -0.01 on the two runs that produced the selection. The cause is a uniform shift of 28 in every gene index. Between the runs the genome gene set went from 6,607 to 6,579 entries, the 28 absent entries sort first, and the cell graph the indices are read from is rebuilt from the live genome at every use. Re-scoring stored triples through the validated path of Section 6.2 reproduces each run at correlation 0.999998 under exactly one index map and not the other.

The symptoms, the decisive test, the second defect that scored panel triples as doubles, the corrected panel predictions and the guard now in place are in the companion document *A Gene Index That Moved* (`notes-tex/010-index-defect`); the additive anchor and the run-to-run agreement tables that exposed the defect are reproduced there from `inference_run_consistency.py`. What matters for this document is what the defect leaves standing. Everything derived from the `inference_2` and `inference_3` prediction files is invalid, which is the panel-12 and panel-24 selections. The `inference_1` artifacts are unaffected, and so is every number in Sections 6 and 6.4: they come from the labeled split through the validated scoring path, whose index space was checked against the recorded metrics before anything was read from it.

7.3 Seed instability in the selection tail, separately ✗

On `inference_1`, where predictions do track gene identity, the three checkpoints still disagree substantially about which triples are extreme. Checkpoint-to-checkpoint Pearson is 0.524 to 0.560, no higher than each checkpoint’s 0.511 to 0.579 agreement with the additive model, and top-100 overlap between two checkpoints is 0.21 to 0.35. Two independent training runs of one architecture on one split nominate largely different top-100 triples, so a selection taken from a single checkpoint’s tail is not reproducible across seeds even where the inference pipeline is sound.

8 Modernizing the 010 build ✗

Two of the obstacles in this document have the same root. Per-record predictions had to be rebuilt by hand rather than recomputed through the stock evaluation path, and the transformer has no row in Table 10, because the December 2025 build no longer loads under the current code.

8.1 What actually broke ✗

Two later changes invalidated it, and neither is about what was measured.

1. The perturbation-ontology refactor of 2026-07-08 renamed the deletion literals. Of the 1,130,196 stored perturbations, 1,008,761 carry `perturbation_type: "deletion"` with `deletion_type: "KanMX"`, which is now `sga_kanmx_deletion`, and 38,742 carry `deletion_type: "mean"`, which is now `mean_deletion`. The remaining 82,693 are allele and temperature-sensitive-allele perturbations and were not renamed. No record in this build is a `natMX` deletion.
2. The component-based media change of 2026-07-14 made `Media.is_synthetic` required. Every record and every reference in the build lacks it.

An earlier version of this document named only the first, and named the wrong replacement vocabulary for it. Both are pure vocabulary changes: neither alters which gene was perturbed, what was measured, or what the label is.

8.2 The migration, and why it copies rather than rebuilds ✗

Every published 010 artifact addresses records by position. The split file, the label table, the perturbed-gene index and the saved per-record predictions are all position-keyed, and the transformer checkpoints were selected against that split. A rebuild from the served graph returns records in a different order, so it would preserve the science and destroy the correspondence. Precedent is on record: carrying 010's splits into experiment 025 by a key that included the perturbation type matched 682 records of 376,732, because the type strings had been renamed in between, while a gene-set key matched all of them.

The migration is therefore a key-preserving copy. It rewrites the two vocabulary fields, writes a sibling build, and leaves the frozen original untouched, which it must, since the original's files are owned by the graph-build user and hardlinked into the graph tree. All 376,732 records migrate, all 376,732 validate against the current schema, and the check that the copy is faithful is that every position still carries the same three systematic gene names and the same label.

One thing the migration deliberately does not fix. The frozen build records YEPD at 30 degrees. The Kuzmin loaders now emit triple-mutant selection medium at 26 degrees, which is the corrected reading of the screen. Applying that here would change the experimental content of the very records whose published results the migration exists to preserve. A build with the corrected environment is a different data version, and it is a rebuild.

8.3 Which experiment ran on which data ✗

The migration creates a second 010 build, so the question of recording which version an experiment used is now concrete rather than hypothetical.

A commit hash does not answer it, for a reason this document supplies. The off-by-28 defect of Section 7.2 happened with no code change at all: the library was untouched between the runs, and what moved was the genome gene set the cell graph is rebuilt from. Two runs at the same commit scored different genes. Identity has to be carried by the built artifact, not by the code that built it.

What already exists is most of the machinery. Dataset builds get a `build_manifest.json` recording a contract fingerprint for every schema symbol the loader depends on, plus build time, host and commit, and staleness is judged by refingerprinting rather than by comparing commits. The gap is that experiment builds are not produced by that class and get no manifest, so neither 010 nor any later experiment build has one.

Three things would close it, in order of how much they are worth.

1. **Pin the index space with the build.** The gene ordering a checkpoint was trained against is the artifact whose drift caused the worst defect here, and it is currently recomputed at every use. Writing it into the build and loading it, rather than reconstructing it, makes the failure impossible instead of merely detected. The guard added in Section 7.2 is the detection half.

2. **Give experiment builds a manifest.** Extend the existing manifest to the experiment-build class, so a build records the schema contract it was made against and any run can name its data version.
3. **Record the source graph’s identity.** A build made by querying the graph database stores the query but nothing about which database answered it. A build identifier from the server, recorded alongside the query, would make the chain complete.

The migrated build carries a `build_provenance.json` naming its source, that source’s LMDB checksum, the exact field rewrites, and what was deliberately not changed. That is the first item’s shape, written by hand for one build.

9 What this establishes, and what it does not ✗

9.1 Established ✗

The additive null had never been fit on 010, and now it has, on the same records, split and label the transformer used, with the ridge penalty chosen on validation and reported on test.

Every model in the comparison receives the same information, the identity of the three perturbed genes. That is a reading of the code rather than an assumption: the encoder is applied at batch size one to a strain-independent input, and the nine interaction graphs reach the prediction only through the weights that training left behind.

The transformer’s margin over a model that cannot represent interaction is real and small on the published split.

The published split was measuring screen identity more than trigenic biology, and this is now measured rather than suspected. A model that never looks at the third gene reaches 97 percent of the additive null’s score, and making query doubles disjoint drops the additive null from 0.400 to 0.127 ± 0.033 and inverts the ordering of the baselines.

The graphs are load-bearing during training. They carried 99.99 percent of the loss, their heads recover their graphs, and the matched pair of runs without the penalty does not train.

The design-space defect has a confirmed cause, a uniform 28-position gene index shift, and the artifacts it invalidates are enumerated.

9.2 Not established ✗

The reproduction is faithful but not bit-exact. The hand-built forward pass matches the recorded metrics to the fourth decimal, not exactly. The original runs used mixed bf16 precision and this one runs in fp32, which is the likely source, but that has not been verified by rerunning in bf16. Conclusions in Section 6.4 rest on correlations at the second decimal and are not sensitive to a fourth-decimal shift.

One seed of the nonlinear baseline enters the paired analysis. Table 9 uses B5 seed 0. The three seeds agree closely on aggregate metrics, but their record-level agreement with each other was not measured, so the B5 column should be read as one draw.

The transformer has not been measured on the disjoint split. Table 10 has no transformer row. Its checkpoints trained on the random split, so scoring them on held-out query pairs would measure memorization. Whether the transformer’s +0.04 margin survives a disjoint split is not measured, in either direction. The run that measures it is queued: job 1609 on GilaHyper trains the replication configuration on the 025 query-pair-disjoint split (Section 6.6), and the baselines it will be read against are already in Table 11.

Whether the graphs carry information is separate from whether they are load-bearing. The runs without the graph penalty fail to train, and the penalty was almost the entire loss. Both facts are recorded. Neither separates the graphs supplying biological structure from a large auxiliary term supplying conditioning that the optimization needs. The control that would separate them, a comparable penalty against degree-matched random graphs, has not been run.

The graph ablation is short and unreplicated in one arm. The matched pair ran 30 epochs against the 49 to 64 of the reported checkpoints, and the arm without the penalty produced two numbers and one diverged run rather than three.

Nothing here bears on the architecture’s merits elsewhere. The reduction in Section 3.4 is specific to the 010 configuration, where the masking and propagation arms are off and the perturbation transform is one block. A configuration that switches those on would have a strain-dependent encoder path and would need its own reading.

9.3 Next

1. Train the transformer on the query-pair-disjoint split and fill the empty row of Table 10. The migrated build makes this runnable, and it is the measurement the rest of this document is now waiting on.
2. Regenerate the panel-12 and panel-24 selections from a correctly indexed inference run. The current selections are invalid, not merely uncertain.
3. Run the graph ablation against degree-matched random graphs, to separate the graphs carrying structure from a large penalty carrying conditioning.
4. Pin the gene index space with the build rather than reconstructing it from a live genome, which is the fix that would have prevented the defect the guard now only detects.
5. Report an ensembled ranking for any future selection rather than one checkpoint's tail, given that two training runs share 0.39 to 0.47 of their top 100.
6. Give experiment builds the manifest that dataset builds already get, so a run can name its data version.